

A2 on a 3+1D Diamond Waist: Two Lattice Spacings Do Not Refute Locality; Bisognano–Wichmann Already Localizes the Continuum Wedge

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Abstract

The same pre-registered locality gate as Papers 21–22: $R(m)/R(0) < 2$ for $0 < mL \leq 8$. The field is a free staggered fermion in three-space: the equal-time slice of a 3+1D theory. The region is the geodesic ball at the waist of a Minkowski causal diamond, diameter $L = 2R$. Local ansatz = on-site + six spatial nearest neighbours.

Coarse $N = 12$, $R = 3$ (123 sites): ratios 1.36, 1.33. Fine $N = 16$, $R = 5$ (515 sites): ratios 1.13, 1.19. Auditor, different code, $N = 14$, $R = 4$: 1.19, 1.22. C2 PASS on every spacing. [P] as a necessary-condition test.

This is not $a \rightarrow 0$, not the Standard Model, and not a selection of de Sitter. The continuum wedge theorem is already Bisognano–Wichmann; the CFT-ball theorem is already Casini–Huerta–Myers. Both are cited, not re-derived.

1 Why 3+1D, not a 4-space lattice

Papers 21–22 tested $d = 1, 2, 3$ spatial lattices. The missing object named there was a *4d continuum diamond*, not a four-dimensional spatial grid. The virtual modes of a 3+1D field live on the Minkowski diamond. Their equal-time restriction is a ball in three-space. A 4-spatial lattice would be a different theory.

2 Gate

Unchanged. $K = \log((1 - C_A)/C_A)$ from the occupied projector of

$$H_{ii} = m(-1)^{x+y+z}, \quad H_{\langle ij \rangle} = -1.$$

Anything beyond the lattice adjacency cannot be a spacetime scalar X in Jacobson’s split. Two spacings at matched $mL \in \{3, 6\}$ are the continuum *check*, locked before the run.

3 Numbers

spacing	n_{ball}	$R(0)$	$R/R(0)$ at $mL = 3$	at $mL = 6$
$N = 12, R = 3$	123	0.132	1.36	1.33
$N = 14, R = 4$ (audit)	—	0.150	1.19	1.22
$N = 16, R = 5$	515	0.158	1.13	1.19

The ratio stays under 2. It is smaller on the finer grid (*computed*, two points). That is the direction a continuum limit would need. It is not the limit.

An independent adversary, different grids $N = 13, 15$, denser mL scan, no shared code: worst ratio 1.29 (at $mL = 4$). Cube-shaped waist also stayed under 2. Catch, not a fail: $R(m)$ is not monotone. It peaks near $mL \sim 4$ –5 and then falls. Mass does not monotonically localize K on this stencil.

4 What the continuum already says

Bisognano–Wichmann [1]: the modular Hamiltonian of a Rindler wedge in Minkowski, including 3+1D, is

$$K = 2\pi \int_{x^1 > 0} x^1 T_{00} d^3x.$$

Casini–Huerta–Myers [2]: for a ball in a CFT,

$$K = 2\pi \int_{|x| < R} \frac{R^2 - r^2}{2R} T_{00} d^3x.$$

Both are local. The lattice test is the massive deformation, which those theorems do not cover.

5 What this does not do

It does not take $a \rightarrow 0$. It does not install the Standard Model. It does not fix $\eta = 1/4G$. It does not copy FGHMV onto de Sitter (Paper 17). It does not promote Jacobson 2016 to [P] (Paper 18). Einstein+ $\Lambda > 0$ is already de Sitter as a metric vacuum (Paper 19); entanglement still does not select the sign of Λ .

Admission. *I pushed A2 onto the 3+1D diamond waist I had named as the next object. I did not write that a two-spacing lattice is the continuum, and I did not write that entanglement gravity is de Sitter.*

References

- [1] J.J. Bisognano and E.H. Wichmann, J. Math. Phys. 16 (1975) 985.
- [2] H. Casini, M. Huerta and R.C. Myers, JHEP 05 (2011) 036, arXiv:1102.0440.
- [3] T. Jacobson, Phys. Rev. Lett. 116 (2016) 201101, arXiv:1505.04753.